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Studierichting: Informatica

Oefeningenreeks 1D nr. 2.3

Bereken a , b en c zodanig dat $a(z^2 + 1) + b(z + i) + c(iz^2 + 2z) = 3i$:

$$\Rightarrow az^2 + a + bz + bi + ciz^2 + 2cz = 3i$$

$$\Rightarrow z^2(a + ci) + z(b + 2c) + (a + bi) = 3i$$

$$\Rightarrow \begin{cases} a + ci = 0 \\ b + 2c = 0 \\ a + bi = 3i \end{cases}$$

$$\Rightarrow \begin{cases} a = -ci \\ b = -2c \\ a = 3i - bi \end{cases}$$

$$\Rightarrow \begin{cases} a = -ci \\ b = -2c \\ -ci = 3i - (-2c)i = 3i + 2ci \end{cases}$$

$$\Rightarrow \begin{cases} a = -ci \\ b = -2c \\ -3ci = 3i \Rightarrow c = 3i/(-3i) = -1 \end{cases}$$

$$\Rightarrow \begin{cases} a = -ci = i \\ b = -2c = 2 \\ c = -1 \end{cases}$$

$$\Rightarrow \boxed{a = i, b = 2, c = -1}$$

References